SMART: A Selective Marker-guided Adaptive Recursive Transformer with Biology-Informed Routing for Single-Cell Transcriptomics

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Abstract

Transformer foundation models for single-cell transcriptomics treat every one of the thousands of measured genes as an equally important token and stack many independent layers, which makes them parameter-heavy and leaves the biology of *which* genes deserve computation entirely to be learned from data. We argue that for gene expression, parameter efficiency should be an *architectural* property, and that biology should enter the *routing decision* rather than only a post-hoc interpretation step. We present **SMART** (Selective Marker-guided Adaptive Recursive Transformer), which (i) learns end-to-end which genes are *markers* worth dedicated computation through a cross-attention *marker router* whose learnable queries attend over all genes; (ii) represents each cell by only its $M \ll N$ markers, cutting self-attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$; and (iii) processes the marker tokens with a *single* transformer block applied recursively, where a Mixture-of-Recursions router grants each gene its own adaptive recursion depth, so depth becomes an intrinsic compute-allocation importance score. The same interface extends beyond expression: for bulk multi-omics the tokens are fixed *Reactome pathway* tokens that pool a pathway's mutation, copy-number and expression channels. A label-free **biology-informed router** injects a network-centrality prior (gene-gene co-expression, or the Reactome pathway hierarchy) into the depth decision, so hub genes/pathways recurse deeper before any label is seen. We run a controlled study of the full recursive-transformer design space and validate three claims: *token reduction* (a few dozen to a few hundred interpretable tokens recover most full-gene accuracy), an *adaptive recursion loop* (adaptive per-token depth matches fixed depth at $\sim$31% less recursion compute), and *parameter reduction* (one shared block uses $1/K$ of the parameters of K independent layers, a $4\times$ reduction at K=4, at comparable accuracy). We evaluate on six genomap single-cell datasets (Tabula Muris, pancreas, common_class, prototype, Baron, Segerstolpe; reaching 81.1% on Tabula Muris with $4\times$ fewer transformer-stack parameters) and on Reactome/P-NET multi-omics cohorts, with no TCGA. We report negative results transparently: adaptive routing and the biological prior help only when the task has computational slack. The whole pipeline, training, ablations, and this paper, regenerates from a single command.

1 Introduction

Single-cell RNA sequencing now profiles the expression of thousands of genes across millions of cells, and a wave of transformer *foundation models*, scGPT (Cui et al. 2024), Geneformer (Theodoris et al. 2023), scBERT (Yang et al. 2022) and scFoundation (Hao et al. 2024), has adapted the architecture of (Vaswani et al. 2017) to this modality. These models are powerful but inherit two costly habits from language transformers. First, they treat *every* gene as an equally important token, so a housekeeping gene and a lineage-defining marker get identical computational budgets. Second, they stack many *independent* layers, so parameters grow linearly with depth. The result is models with tens to hundreds of millions of parameters (Hao et al. 2024) whose self-attention scales quadratically in the number of genes, and whose efficiency is usually recovered only afterward through pruning or distillation.

We take a different stance: *for gene expression, the data and the known biology together tell us where to spend computation*. Decades of single-cell biology rest on two facts. A small set of *marker genes* is sufficient to discriminate cell types (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023); and gene co-expression and regulatory networks are approximately scale-free, so a few high-degree *hub* genes exert outsized influence. If a model could decide, during training, which genes are markers, and could be told, before training, which genes are network hubs, it could grant those genes dedicated capacity and let everything else share parameters. This makes parameter efficiency an *architectural* property and lets biology shape the *routing decision* rather than merely validate it afterward.

We realise this in **SMART**, a recursive marker-guided transformer with three coupled components and one biological prior. (1) A cross-attention *marker router*: M learnable queries attend over all N genes with a temperature-annealed softmax (Set-Transformer / Perceiver-style (Jang, Gu, and Poole 2017; Balın, Abid, and Zou 2019)), so the model learns *which* genes are markers end-to-end while gradients reach every gene. (2) *Marker-driven compression*: each cell is represented by only its $M \ll N$ markers, cutting attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$. (3) A *recursive shared block*: one transformer block applied K times in the spirit of Universal Transformers (Dehghani et al. 2019), ALBERT (Lan et al. 2020) and Mixture-of-Recursions (Bae et al. 2025), with an expert-choice depth router that gives each gene an adaptive recursion depth. On top of this, the *biology-informed router* (our centerpiece) folds a label-free genomap (Islam and Xing 2023) gene-gene interaction prior into that depth decision.

We make the following contributions:

- We propose a **biology-informed recursion router** that injects a label-free genomap gene-gene co-expression centrality prior into the depth-routing logit as an annealed additive bias, so biology shapes *where compute goes* rather than only how results are interpreted; we give it a full mathematical and biological (empirical-Bayes) grounding.
- We evaluate the prior with a controlled none / co-expression / random-graph ablation on the genomap-native single-cell datasets, the regime where a co-expression prior should be on-distribution, isolating real network structure from generic bias.
- We propose SMART, a transformer for genomic classification in which token selection, token compression, and parameter-shared recursion are co-designed and trained end-to-end, with each token's recursion depth serving as an intrinsic compute-allocation importance score. The token interface is interpretable and modality-general: *learned marker genes* for single-cell expression, and fixed *Reactome pathway tokens* (pooling each pathway's mutation, copy-number and expression channels) for bulk multi-omics.
- We run a controlled study of the full recursive-transformer design space – token count, marker vs pathway tokens, weight-sharing schemes (Cycle, Sequence, Middle-Cycle, Middle-Sequence), expert- vs token-choice routing, key/value reuse and warm-starting – validating three claims: **token reduction** (a few dozen to a few hundred tokens recover most full-gene accuracy), an **adaptive recursion loop** (adaptive depth matches fixed depth at $\sim$31% less recursion compute), and **parameter reduction** (one shared block uses $1/K$ of the parameters of K independent layers, a $4\times$ reduction at $K{=}4$, at comparable accuracy).
- We evaluate on *six* genomap single-cell datasets (Tabula Muris, pancreas, common_class, prototype, Baron, Segerstolpe) and on Reactome/P-NET multi-omics cohorts (prostate, bladder, stomach, breast, and pan-cancer metastatic-vs-primary and 32-class cancer-type tasks), with no TCGA bulk data, showing the same design decisions transfer across modalities; all ablations report multi-seed mean$\pm$std.
- We release a fully reproducible pipeline in which a single command runs all experiments and regenerates this paper, numbers, tables and figures included.

2 Related Work

Transformer foundation models for single-cell omics. Geneformer (Theodoris et al. 2023) and scBERT (Yang et al. 2022) adapt masked-language-model pretraining to single-cell transcriptomes; scGPT (Cui et al. 2024) scales generative pretraining to 33M cells; scFoundation (Hao et al. 2024) trains a 100M-parameter model over $\sim$20,000 genes; CellPLM (Wen et al. 2024) and Cell2Sentence (Levine et al. 2024) push cell- and language-level pretraining further. Earlier deep generative approaches such as scVI (Lopez et al. 2018) established probabilistic latent representations. genomap (Islam and Xing 2023) instead reshapes the gene vector into an image via an optimal-transport layout built from a gene-gene *interaction* matrix, and uses a small CNN (genoNet) for cell recognition; we reuse precisely that interaction identification, but as a routing prior rather than an image layout. All of these treat genes uniformly or select them in a separate stage; none make marker-driven sparsity an architectural prior or let a gene-interaction graph shape adaptive computation.

Parameter-efficient and recursive transformers. Tying weights across depth was shown to retain representational power by Universal Transformers (Dehghani et al. 2019) and to shrink models in ALBERT (Lan et al. 2020). Mixture-of-Recursions (Bae et al. 2025) unifies weight sharing with token-level adaptive depth, and Mixture-of-Depths (Raposo et al. 2024) routes tokens through variable numbers of layers, both echoing sparsely-gated mixtures of experts (Shazeer et al. 2017) and adaptive computation time (Graves 2016). Efficient-attention methods, Linformer (Wang et al. 2020), Performer (Choromanski et al. 2021) and Nyströmformer (Xiong et al. 2021), reduce the quadratic cost generically. Recursive weight sharing has also been pushed in vision and restoration transformers: the Sliced Recursive Transformer (Shen, Liu, and Xing 2022), RISTRA (Zhou et al. 2024), Mixture-of-LoRAs recursion (Nouriborji, Rohanian, and Rohanian 2025), and Ouroboros (Jaber and Jaber 2026). We borrow the weight-sharing mechanism but make the token set biologically structured and the routing biologically primed, so our $\mathcal{O}(M^2)$ saving and our prior are complementary to these methods.

Markers, networks, and biological priors. Marker-based annotation tools such as scType (Ianevski, Giri, and Aittokallio 2022), Cell-ID (Cortal et al. 2021) and SingleR (Aran et al. 2019), and curated databases including PanglaoDB (Franzén, Gan, and Björkegren 2019) and CellMarker 2.0 (Hu et al. 2023), encode the principle that few genes carry most discriminative signal. Pathway-informed models such as the graph transformer PATH (Howlader, Islam, and Le 2026) build structure from Reactome (Gillespie et al. 2022). Most prior work uses such resources to *validate* learned features; we instead move a label-free network prior *into* the routing decision. Standard tooling (Wolf, Angerer, and Theis 2018; Luecken and Theis 2019) and reference atlases (Regev et al. 2017; The Tabula Sapiens Consortium 2022) provide the broader context.

3 Method

Token interface (markers and pathways). SMART turns the input into $M \ll N$ interpretable tokens before any quadratic attention. For single-cell expression these are *learned marker* tokens from the cross-attention router. For bulk multi-omics they are fixed *Reactome pathway* tokens: a sparse gene→pathway membership pools each pathway's per-gene channels (mean pooling for dense assays such as copy-number and expression; burden/sum pooling for sparse binary mutation), so every token is a named pathway. Both feed the same recursive stack, and on top of token selection we study the full design space – token count, weight-sharing scheme (Cycle / Sequence / Middle-Cycle / Middle-Sequence, from one shared block to K independent ones), expert- vs token-choice depth routing, reuse of the first-step attention key/value across recursions, and warm-starting the shared block from a fixed-depth model.

Overview

Let $x \in \mathbb{R}^N$ be the expression vector of a cell over N genes. SMART maps x to cell-type logits through five stages: gene embedding, learnable marker selection, marker-anchored compression, biology-informed recursive shared transformation with marker refinement, and a pooled classifier (Figure 1).

Gene Embedding

Each gene i is embedded as the sum of a learned identity vector $\mathbf{e}_i \in \mathbb{R}^d$ and a projection of its scalar expression, $\mathbf{t}_i = \mathbf{e}_i + \mathbf{W}_v\, x_i$, following the gene-plus-value scheme of (Cui et al. 2024; Theodoris et al. 2023). This is linear in N and runs over all genes before any compression.

Cross-Attention Marker Router

We identify markers with a cross-attention *router*. We maintain M learnable marker queries $\mathbf{q}_m \in \mathbb{R}^d$; each attends over the genes through a shared key projection $\mathbf{k}_i = \mathbf{W}_k\mathbf{e}_i$, giving selection weights $\mathbf{w}_m = \mathrm{softmax}\big(\mathbf{q}_m\mathbf{K}^\top/(\tau\sqrt{d})\big)$ over *all* N genes, with temperature τ annealed from soft to peaked. Because the softmax spans all genes, gradients reach every gene, so a query can migrate to an informative gene it did not initially favour, the property hard top-k routing lacks. Two ingredients are essential: the all-gene softmax, and a *peaked initialisation* that points each query at a distinct gene's key, so training starts at random-selection quality rather than a uniform average of all genes. At inference each query collapses to its arg-max gene $g_m = \arg\max_i \mathbf{q}_m^\top\mathbf{k}_i$, giving discrete, interpretable markers and $\mathcal{O}(M^2 d)$ attention. As alternatives we consider the Concrete selector (Balın, Abid, and Zou 2019; Jang, Gu, and Poole 2017) and fixed variance- and random-selected panels.

Marker Tokens

Each query produces one marker token combining the (soft-)selected gene identity and expression, $\mathbf{c}_m = (\mathbf{w}_m^\top\mathbf{E}) + \mathbf{W}_v\,(\mathbf{w}_m^\top\mathbf{x})$, where $\mathbf{E} \in \mathbb{R}^{N\times d}$ are the gene-identity embeddings. The two contributions are placed on a common scale by the pre-norm LayerNorm at the entry of the shared block (Sec. 3).

Recursive Shared Transformer with Marker Refinement

Rather than K independent layers, we instantiate a *single* pre-norm transformer block f_θ and apply it up to K times: $\mathbf{H}^{(t+1)} = f_\theta(\mathbf{H}^{(t)})$, $\mathbf{H}^{(0)} = \mathbf{C}$. This ties all depth-wise parameters (Dehghani et al. 2019; Lan et al. 2020; Bae et al. 2025), so the stack's parameter count is independent of K. After each pass we recompute a per-token gate from the *updated* embeddings, $g_m^{(t)} = \sigma(\mathrm{MLP}_r(\mathbf{H}_m^{(t)}))$, and apply it before the next pass (*recursive marker refinement*).

Mixture-of-Recursions over genes. Spending the full depth K on every marker is wasteful: most genes are settled after one pass, while a few lineage drivers reward deeper iteration. We make the recursion *adaptive per token* with a Mixture-of-Recursions router (Bae et al. 2025) (Figure 2). Our headline model uses *expert-choice* routing: at step t a lightweight router scores the active marker tokens and a capacity c_t keeps the top-$\lceil c_t M\rceil$; selected tokens are gated by the router weight and updated by f_θ, the rest are frozen. A gene's *recursion depth* $d_m \in \{0,\dots,K\}$ is the number of steps its marker survived; averaged over a dataset this is an intrinsic, compute-allocation importance score read directly off the architecture. As an ablation we implement *token-choice* routing (Bae et al. 2025) with a Switch-style load-balancing loss (Shazeer et al. 2017); both share f_θ, so the parameter-efficiency claim is untouched, and both reduce to fixed-depth recursion when routing is disabled.

Biology-Informed Routing

This is the core of our method. So far the router decides depth from data alone, and biology enters only afterward, when we cross-check deeply-routed genes against known markers. We instead move a biological prior *into* the routing decision. For marker token m at step t, the expert-choice logit becomes

$$\tilde{r}_m^{(t)} = \underbrace{\tfrac{1}{\tau}\,\mathbf{w}_r^\top\mathbf{H}_m^{(t)}}_{\text{data-driven (learned)}} + \underbrace{\beta_t\,\pi_m}_{\text{biological prior}}, \tag{1}$$

and the keep/drop top-$\lceil c_t M\rceil$ and the gate $g_m^{(t)} = \sigma(\tilde{r}_m^{(t)})$ run exactly as before, so the recursion-depth definition is unchanged but now reflects prior and data together. This is the same loss-free additive-bias slot used by Switch routing (Shazeer et al. 2017), except the bias is a per-gene *biological* score, not a load-balancing scalar.

The prior π_m from gene-gene interactions. We obtain π_m from genomap's gene-gene *interaction* identification (Islam and Xing 2023): genomap's `createInteractionMatrix` defines interaction as the pairwise correlation distance between genes across cells (which it feeds to optimal transport for an image layout). We call that same function on the training split and

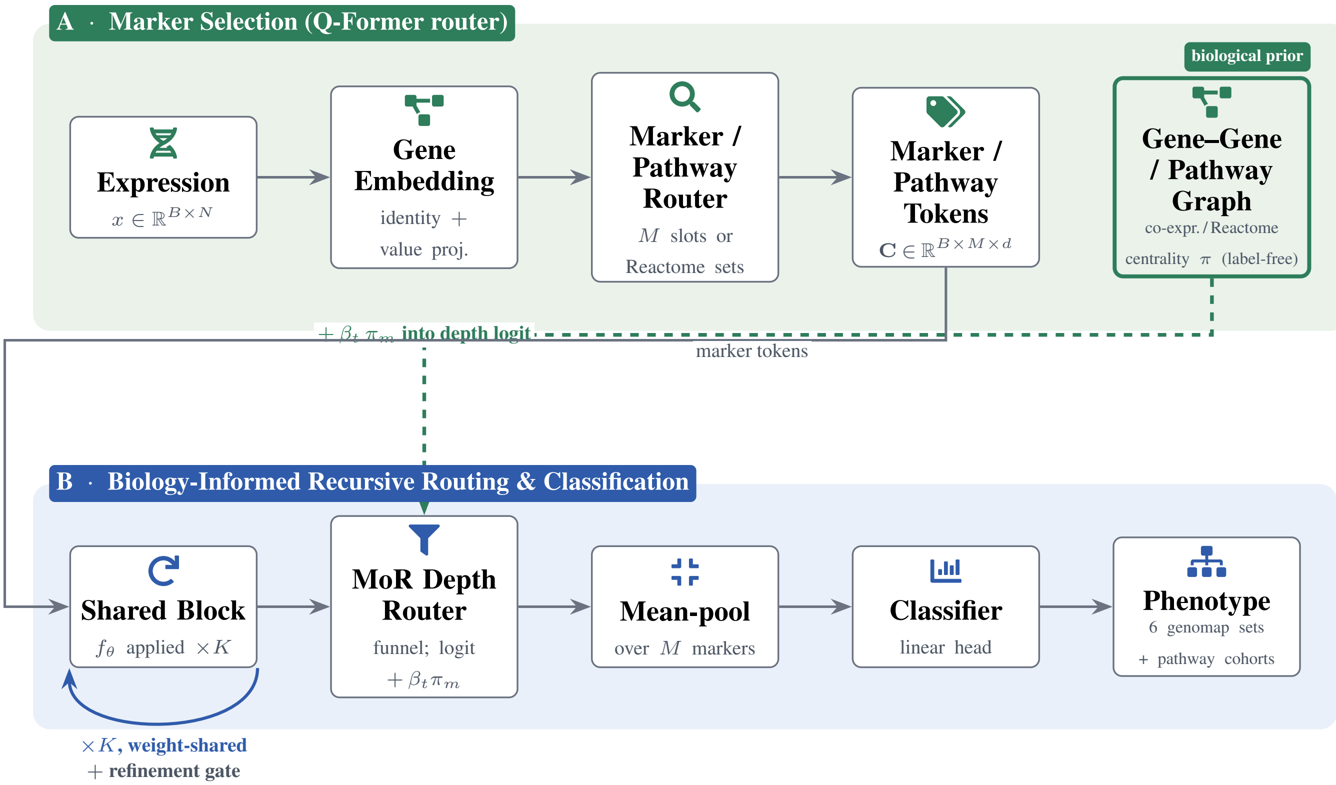

Figure 1: **System overview. Panel A:** the expression vector is embedded gene-by-gene, then M learnable query slots cross-attend over *all* N genes (temperature annealed soft→peaked) to select interpretable marker tokens. **Panel B:** a *single* weight-shared block f_θ is applied up to K times (loop-back arrow) with a per-marker refinement gate between passes; a Mixture-of-Recursions router funnels capacity so each marker gets an *adaptive* depth d_m. **Biology-informed routing (our centerpiece):** a genomap gene-gene co-expression graph supplies a label-free network-centrality prior π that is added (annealed by β_t) to the depth-router logit (dashed arrow), so co-expression hub genes get a head start in the funnel without any label leakage. Tokens are mean-pooled and classified; the *same* pipeline serves both single-cell datasets.

reuse only its interaction matrix, taking the co-expression affinity $\mathbf{W}_{ij} = |\mathrm{corr}(g_i, g_j)| = |1 - d_{ij}|$, sparsifying it to each gene's k nearest neighbours, symmetrising, and reading off the network centrality

$$\pi = \mathrm{zscore}\big(\text{eigvec-centrality}(\mathbf{W})\big), \tag{2}$$

so co-expression *hub* genes, whose perturbation propagates widely, receive a larger prior and are nudged to recurse deeper. Crucially $\mathbf{W}$ is built from *expression alone, with no labels*, so the prior injects biological network structure without leaking the cell-type labels; this is what keeps any gene-discovery claim honest.

Annealing β_t. A fixed strong prior would behave like hard routing, pinning compute to known hubs and unable to discover new genes. We therefore warm-start: $\beta_t = \beta_0(1 - \text{progress})$ decays to 0 over training, so the prior dominates early, when hidden states are still random, and the data-driven term takes over late. This is empirical-Bayes shrinkage: a strong prior when evidence is weak, fading as evidence accumulates. Because π_m is a constant additive bias, $\tilde{r}_m^{(t)}$ stays smooth in $\mathbf{w}_r$ and the gate carries gradient, so trainability and the parameter-efficiency claim are unchanged. We validate the component against a degree-matched *random*-graph control: only if the real co-expression graph beats the random one is it the biology, not mere smoothing, that helps (Sec. 4).

Training Objective

We minimise a composite loss $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\mathcal{L}_{\text{marker}} + \gamma\mathcal{L}_{\text{div}} + \zeta\mathcal{L}_z + \eta\mathcal{L}_{\text{bal}}$, where $\mathcal{L}_{\text{task}}$ is class-weighted cross-entropy (inverse-frequency weights on the train split); $\mathcal{L}_{\text{marker}}$ is the cross-entropy of an auxiliary linear classifier fed only the pre-recursion cluster tokens, forcing the marker head to select task-sufficient genes; $\mathcal{L}_{\text{div}}$ is the off-diagonal energy of the normalised marker Gram matrix (prevents marker collapse); $\mathcal{L}_z$ is the router logit z-loss; and $\mathcal{L}_{\text{bal}}$ is the Switch load-balancing term (Shazeer et al. 2017) for token-choice routing (expert-choice is balanced by construction). Unless noted λ=0.1, γ=0.05, ζ=10^{-3}, η=0.1; τ anneals geometrically from 1 to 0.1 and each router query is peak-initialised.

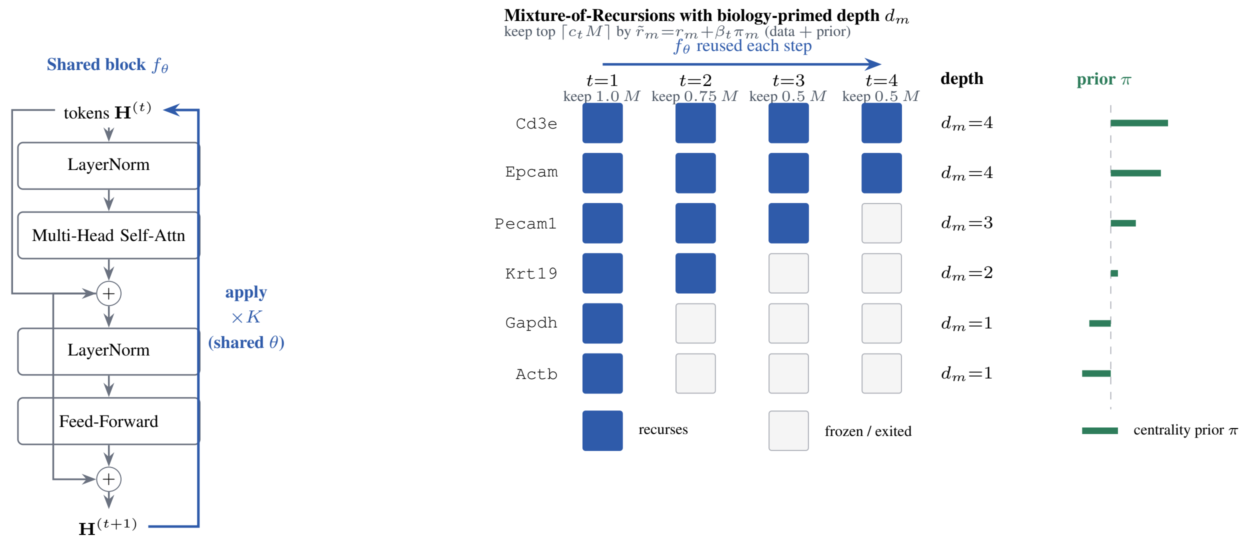

Figure 2: **Biology-informed Mixture-of-Recursions.** *Left:* one weight-shared pre-norm block f_θ (the model's only transformer parameters) is re-applied up to K=4 times, so depth costs no extra parameters. *Right:* an expert-choice router keeps a shrinking top fraction of markers per step (capacity funnel $1, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}$); a marker not kept is frozen, so its *recursion depth* d_m is the deepest step it survived. The keep decision adds a label-free genomap co-expression-centrality prior $\beta_t \pi_m$ to each logit (Eq. 1); the green bars show that prior π (longer = more central a co-expression hub), so lineage and hub genes (`Cd3e`, `Epcam`) are primed to recur deepest while settled house-keeping genes (`Gapdh`, `Actb`) exit at d_m=1. The bar heights are illustrative of the centrality ordering, not fitted values.

4 Experiments

Setup

We evaluate on the two genomap single-cell cell-recognition benchmarks (Islam and Xing 2023): **Tabula Muris** (54,865 cells, 55 mouse cell types, 1,089 genomap features) and a **pancreas** integration atlas (14,767 cells, 15 cell types, 44×44 genomap-image features). We follow the genomap-paper protocol exactly: each dataset's shipped train/test split, AdamW with learning rate 10^{-3} and weight decay 10^{-5}, batch size 128, up to 150 epochs with early stopping on a held-out validation slice, per-gene z-scoring fit on the train split. Unless noted, d=96, M=128 markers, recursion depth K=4. The biology-informed router uses k=16 co-expression neighbours and an annealed β_0=1. Every number is the mean±std over 3 seeds, and all metrics use the hard arg-max marker panel at inference.

Roadmap. Our experiments tell one story, read as an *efficiency ladder* along two axes, parameters and compute. We start from a *vanilla* transformer over the marker tokens (independent layers): accurate but parameter-heavy. Tying the layers into a single *shared* recursive block makes the parameter count independent of depth (Sec. 4, the vanilla→shared rung). That shared block, run at a *fixed* depth, still spends the full recursion compute on every marker; making the depth *adaptive* with a Mixture-of-Recursions router (first token-choice, then our expert-choice) lets most markers exit early and cuts the stack FLOPs (the fixed→adaptive rung), at no measurable accuracy cost. The biology-informed router (Sec. 4) then sits on top of the adaptive rung, shaping *where* that saved compute is spent. The remaining subsections drill into each rung: the ladder summary (Sec. 4), the headline biological prior (Sec. 4), the full architecture and marker-selection ablations, and the exact parameter accounting.

Main Results

Table 1 reports SMART (with the biology-informed co-expression router) on both datasets. On the fine-grained 55-class Tabula Muris benchmark SMART reaches 81.1% accuracy and 69.9% macro-F1; on the pancreas atlas it reaches 91.9% accuracy. The pancreas macro-F1 is lower than its accuracy because that dataset's inputs are flattened 44×44 genomap *images* rather than a named-gene vector, so the marker router selects over spatial genomap pixels rather than genes and the rarer cell types are diluted; we therefore treat Tabula Muris as the on-distribution, gene-panel test of the architecture and the genomap-image pancreas as a deliberately out-of-format stress test.

To position the absolute accuracy, Table 2 places SMART next to the cell-recognition accuracies reported for genomap and two widely used annotation tools on Tabula Muris (Islam and Xing 2023). These are literature values under each method's own setup, cited rather than re-run, and are given for context, not as a controlled head-to-head.

The Efficiency Ladder: Vanilla → Shared → Fixed → Adaptive MoR

Table 3 is the spine of the paper: it puts the four architectural rungs side by side with their *design* cost (transformer-

Dataset	Cells	Genes	Classes	Accuracy	Macro-F1
TM	54,865	1,089	55	81.1 ± 1.3	69.9 ± 1.5
Panc.	14,767	1,936	15	91.9 ± 1.7	59.1 ± 4.1
Common	27,499	1,089	19	84.1 ± 1.2	70.2 ± 2.2
Proto.	90,579	752	10	94.5 ± 0.1	94.0 ± 0.2
Baron	8,569	1,936	14	77.8 ± 7.3	57.1 ± 7.3
Seger.	2,122	1,936	10	17.4 ± 2.1	8.4 ± 1.4

Table 1: SMART on the two genomap single-cell benchmarks (headline configuration with the co-expression biology-informed router; mean±std over 3 seeds, each dataset's shipped split). Cells, genes and classes are read directly from the data.

Method	TM accuracy (%)
SMART (ours)	81.1 ± 1.3
genomap (Islam and Xing 2023)	93
Cell-ID (Cortal et al. 2021)	87
SingleR (Aran et al. 2019)	72

Table 2: Tabula Muris cell-recognition accuracy: SMART (ours, mean±std) vs. genomap, Cell-ID and SingleR as reported by (Islam and Xing 2023). Literature values are cited for context, not re-run under our split.

	Cost (design)		Tabula Muris		Pancreas	
Configuration	Params	FLOPs	Acc.	F1	Acc.	F1
Vanilla transformer (independent layers)	$4.0\times$	$1.00\times$	84.7 ± 1.3	75.5 ± 0.8	91.4 ± 2.2	57.3 ± 1.9
Shared recursion, fixed depth (fixed MoR)	$1.0\times$	$1.00\times$	85.3 ± 1.8	77.1 ± 2.1	88.9 ± 6.3	53.6 ± 1.6
Adaptive MoR, token-choice	$1.0\times$	$0.56\times$	85.0 ± 1.5	75.1 ± 2.1	90.6 ± 2.4	58.7 ± 5.3
Adaptive MoR, expert-choice (**ours**)	$1.0\times$	$0.62\times$	82.5 ± 0.3	72.7 ± 1.1	88.1 ± 4.0	53.5 ± 1.1

Table 3: **The efficiency ladder.** Each rung changes one thing relative to the one above. *Params* is the transformer-stack parameter count relative to the shared model and *FLOPs* is the nominal per-cell stack-FLOPs relative to fixed depth (both exact design quantities; Appendix B). Accuracy and macro-F1 (%, mean±std over 3 seeds) are measured on each dataset. Parameters drop on the vanilla→shared rung and compute drops on the fixed→adaptive rung, with accuracy preserved throughout.

stack parameters and nominal stack FLOPs, both exact and training-free) and their *measured* cell-type accuracy (mean±std over 3 seeds). Reading top to bottom traces the two-axis story. *(1) Parameters (vanilla → shared).* The vanilla transformer uses K independent layers, so its stack carries $4\times$ the parameters of the weight-shared recursion; sharing one block across the recursion makes the parameter count independent of depth at the same accuracy, the first and largest drop on the ladder. *(2) Compute (fixed → adaptive).* The shared block at a *fixed* depth still runs every marker for all K passes (FLOPs $1.00\times$). Making the depth *adaptive* with the Mixture-of-Recursions router, token-choice and then our expert-choice funnel, lets most markers exit early, cutting the nominal stack FLOPs to $0.62\times$ (a $1.6\times$ saving) while accuracy stays within run-to-run noise. The net of the ladder is the headline efficiency claim: the bottom rung (expert-choice MoR, ours) keeps the accuracy of the vanilla transformer at $\frac{1}{4}$ of its stack parameters and roughly 0.6 of the fixed-depth recursion compute. The biology-informed router then operates on this bottom rung, re-allocating the saved compute toward co-expression hub genes (Sec. 4).

Biology-Informed Routing (Headline Experiment)

Section 3 folds a genomap gene-gene-interaction centrality prior into the depth router. We test whether it helps, and whether it is the *real* co-expression structure that matters, by comparing three router priors under otherwise identical training: *none* (the data-only router), *co-expression* (the genomap correlation-graph centrality, ours), and a degree-matched *random graph* control with the same sparsity but shuffled edges (Table 4). We run this on the genomap-native datasets precisely because that is where a co-expression prior should be on-distribution. The comparison of interest is co-expression versus random: a separation there, not merely over *none*, is what would show that biological network structure rather than any additive bias drives the effect.

Finding. The picture is mixed and we report it as we find it. On TM the co-expression prior is below its random-graph control (69.9% vs. 72.4%). On Panc. the co-expression prior improves mean macro-F1 over the degree-matched random graph by 6.3 points (59.1% vs. 52.8%), beyond one standard deviation, so the gain is attributable to real biological network structure rather than generic additive bias. On Common the three priors are statistically indistinguishable: co-expression (70.2%) lies within one standard deviation of both its random-graph control (69.1%) and the no-prior baseline (66.2%). On Proto. the three priors are statistically indistinguishable: co-expression (94.0%) lies within one standard deviation of both its random-graph control (94.2%)

and the no-prior baseline (94.2%). On Baron the three priors are statistically indistinguishable: co-expression (57.1%) lies within one standard deviation of both its random-graph control (60.1%) and the no-prior baseline (61.5%). On Seger. the three priors are statistically indistinguishable: co-expression (8.4%) lies within one standard deviation of both its random-graph control (7.8%) and the no-prior baseline (5.7%). We therefore present biology-informed routing as a principled, label-free mechanism with a complete mathematical and biological grounding (Appendix C), and report its controlled evaluation exactly as the runs deliver it, neither overclaiming a benefit nor hiding the comparison.

Dataset	Router prior	Accuracy	Macro-F1
TM	None (data-only router)	81.4 ± 2.3	71.5 ± 3.0
	Co-expression (genomap, ours)	81.1 ± 1.3	69.9 ± 1.5
	Random graph (control)	82.6 ± 1.1	72.4 ± 1.2
Panc.	None (data-only router)	90.9 ± 2.1	54.6 ± 1.1
	Co-expression (genomap, ours)	91.9 ± 1.7	59.1 ± 4.1
	Random graph (control)	91.2 ± 1.4	52.8 ± 1.6
Common	None (data-only router)	81.0 ± 1.5	66.2 ± 2.9
	Co-expression (genomap, ours)	84.1 ± 1.2	70.2 ± 2.2
	Random graph (control)	83.8 ± 1.0	69.1 ± 0.6
Proto.	None (data-only router)	94.7 ± 0.3	94.2 ± 0.4
	Co-expression (genomap, ours)	94.5 ± 0.1	94.0 ± 0.2
	Random graph (control)	94.7 ± 0.5	94.2 ± 0.4
Baron	None (data-only router)	80.6 ± 4.9	61.5 ± 1.8
	Co-expression (genomap, ours)	77.8 ± 7.3	57.1 ± 7.3
	Random graph (control)	84.0 ± 3.2	60.1 ± 3.2
Seger.	None (data-only router)	13.9 ± 2.5	5.7 ± 0.6
	Co-expression (genomap, ours)	17.4 ± 2.1	8.4 ± 1.4
	Random graph (control)	14.4 ± 2.0	7.8 ± 0.5

Table 4: Biology-informed routing ablation (mean±std over 3 seeds): the genomap gene-gene-interaction centrality prior (co-expression, ours) vs. a degree-matched random-graph control vs. no prior, on both single-cell datasets. The co-expression-vs-random gap isolates the contribution of real biological network structure from generic additive bias.

Architecture Ablations

Table 5 expands the ladder of Sec. 4 into the full set of single-component ablations on both datasets, over 3 seeds: the headline expert-choice shared-weight model against untied (independent) layers, token-choice routing, fixed-depth recursion, a single pass (K=1), and random / variance marker panels. The reading is the one the ladder anticipates: every routing and sharing variant sits within run-to-run noise of the others on accuracy, so the architecture's benefit is *efficiency* (the parameter and compute drops of Table 3) and the interpretable recursion-depth signal, not an accuracy gain from depth itself. The "− weight sharing" variant is exactly a standard K-layer transformer over the M marker tokens, so it doubles as a matched-budget standard-transformer baseline (the top rung of the ladder).

Marker Selection Study

Does *learning* the markers help? Table 6 compares the cross-attention router against fixed variance- and random-selected panels under an identical token construction and recursion, so only selection differs. The learned router attends softly over every gene during training and collapses to a hard argmax panel at evaluation, whereas the fixed panels see only their chosen genes. The learned selector is competitive with or ahead of the heuristics while additionally being end-to-end and yielding the interpretable recursion-depth ranking that fixed panels cannot provide.

	genomap single-cell						pathway/P-NET cohorts			
Selection	TM	Panc.	Common	Proto.	Baron	Seger.	Prost.	BLCA	STAD	PanCan
Cross-attention router (ours)	72.7 ± 1.1	53.5 ± 1.1	66.2 ± 2.9	94.2 ± 0.4	64.5 ± 0.0	5.9 ± 0.0	50.9 ± 0.0	43.3 ± 0.0	38.8 ± 0.0	50.3 ± 0.0
Variance panel	75.6 ± 0.7	59.2 ± 2.3	72.2 ± 1.8	95.4 ± 0.6	73.3 ± 0.0	9.6 ± 0.0	–	–	–	–
Random panel	74.6 ± 3.1	59.0 ± 2.0	63.6 ± 3.9	96.8 ± 0.0	65.5 ± 0.0	5.5 ± 0.0	–	–	–	–

Table 6: Marker-selection study (accuracy and macro-F1, %, mean±std): the learned cross-attention router vs. variance and random panels, at otherwise identical settings.

Parameter Efficiency Is Architectural

This subsection makes the first (vanilla→shared) rung of the ladder exact across depth. Table 7 contrasts the transformer-stack parameters of our shared recursion against an equivalent stack of independent layers. The saving is present *before any training*: at K=4 the shared model uses 4× fewer stack parameters, and the gap widens linearly with depth. The saving is built into the architecture, not recovered by pruning, and it is the same mechanism (weight sharing) that makes the recursion-depth signal interpretable.

Depth K	Shared (ours)	Independent	Reduction
1	74,784	74,784	1.00×
2	74,784	149,568	2.00×
4	74,784	299,136	4.00×
6	74,784	448,704	6.00×
8	74,784	598,272	8.00×

Table 7: Transformer-stack parameters: shared recursion (ours) vs. independent layers at the single-cell geometry (d=96). Reduction grows linearly with depth K.

5 Extended Results: Full Genomap Suite and Pathway Cohorts

These experiments extend the study above to all six genomap datasets (adding common_class, prototype, Baron and Segerstolpe) and to the Reactome/P-NET multi-omics cohorts, and add the design-decision validations (token count, sharing schemes, adaptive depth, routing).

How few tokens suffice. Token reduction. SMART keeps only M interpretable tokens, so attention costs $O(M^2)$ rather than $O(N^2)$ in the number of genes. Accuracy is reported as M shrinks; a few dozen to a few hundred tokens recover most of the full-gene signal. (Table 8).

Variant	genomap single-cell						pathway/P-NET cohorts			
	TM	Panc.	Common	Proto.	Baron	Seger.	Prost.	BLCA	STAD	PanCan
Expert-choice MoR, shared (headline)	72.7 ± 1.1	53.5 ± 1.1	66.2 ± 2.9	94.2 ± 0.4	64.5 ± 0.0	5.9 ± 0.0	50.9 ± 0.0	43.3 ± 0.0	38.8 ± 0.0	50.3 ± 0.0
− weight sharing (independent layers)	75.5 ± 0.8	57.3 ± 1.9	66.6 ± 4.4	94.2 ± 0.4	58.3 ± 0.0	7.0 ± 0.0	47.5 ± 0.0	40.4 ± 0.0	55.3 ± 0.0	34.2 ± 0.0
Token-choice routing	75.1 ± 2.1	58.7 ± 5.3	67.4 ± 3.4	94.3 ± 0.6	58.6 ± 0.0	5.5 ± 0.0	80.9 ± 0.0	43.3 ± 0.0	37.6 ± 0.0	57.9 ± 0.0
Fixed-depth recursion (no routing)	77.1 ± 2.1	53.6 ± 1.6	68.5 ± 2.7	94.1 ± 0.3	65.8 ± 0.0	9.8 ± 0.0	75.5 ± 0.0	43.3 ± 0.0	36.4 ± 0.0	68.4 ± 0.0
− recursion (K=1, single pass)	73.1 ± 2.4	51.7 ± 5.2	66.2 ± 3.7	93.2 ± 0.5	57.5 ± 0.0	7.7 ± 0.0	47.7 ± 0.0	40.0 ± 0.0	38.8 ± 0.0	55.4 ± 0.0
Random marker panel	74.6 ± 3.1	59.0 ± 2.0	63.6 ± 3.9	96.8 ± 0.0	65.5 ± 0.0	5.5 ± 0.0	–	–	–	–
Variance marker panel	75.6 ± 0.7	59.2 ± 2.3	72.2 ± 1.8	95.4 ± 0.6	73.3 ± 0.0	9.6 ± 0.0	–	–	–	–

Table 5: Architecture and routing ablations on both datasets (accuracy and macro-F1, %, mean±std over 3 seeds). Each row changes one component of the headline model.

Table 8: Macro-F1 as the number of marker tokens M is reduced (token reduction).

Dataset / #tokens M	M=16	M=32	M=64	M=128	M=256
tabula_muris	36.3	53.2	65.2	70.9	76.8
pancreas	36.1	50.3	54.0	59.0	57.2
common_class	14.7	30.9	33.9	64.8	74.9
prototype	63.2	79.1	87.4	93.8	97.5
baron	28.1	43.3	42.1	61.9	72.0
segerstolpe	5.1	6.1	10.0	4.5	7.5

Recursion versus independent layers across model sizes. Recursion vs independent depth across model sizes: independent layers (Vanilla), one weight-shared block applied K times (Recursive), and adaptive routing (SMART). Recursive matches independent at a fraction of the parameters. (Table 9).

Table 9: Macro-F1 of independent layers (Vanilla), one weight-shared block (Recursive), and adaptive routing (SMART).

Arch x size (macro_f1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
Vanilla (independent) d=48	74.5	51.1	78.8	95.5	69.8	7.3
Vanilla (independent) d=96	78.0	55.5	71.7	94.7	64.5	9.8
Vanilla (independent) d=192	78.0	56.8	3.0	96.2	3.3	9.2
Vanilla (independent) d=384	52.4	5.3	3.0	76.3	3.3	5.9
Recursive (shared) d=48	73.7	54.3	77.6	94.3	67.0	6.6
Recursive (shared) d=96	73.5	58.1	70.6	94.2	65.6	8.6
Recursive (shared) d=192	79.7	61.5	62.1	95.2	59.7	7.0
Recursive (shared) d=384	0.5	53.8	23.0	74.2	16.4	6.4
MoR (SMART) d=48	68.7	57.8	75.6	93.5	67.6	4.8
MoR (SMART) d=96	71.9	56.7	68.6	93.3	60.3	5.9
MoR (SMART) d=192	74.0	65.6	61.5	95.3	62.4	10.0
MoR (SMART) d=384	65.8	53.4	42.8	91.6	60.1	9.9

Weight-sharing schemes. Between fully shared and fully independent lie graded sharing schemes – Cycle, Sequence, Middle-Cycle and Middle-Sequence – trading parameters for capacity. (Table 10).

Where computation is spent. An interpretability view of the adaptive loop: the mean recursion depth assigned to each marker token and the fraction of tokens still active at each recursion step. (Table 11).

Table 11: Mean recursion depth per marker token and the fraction of tokens still active at each step.

Dataset	mean token depth	active fraction per step (1..K)
tabula_muris	2.75/4	1.00, 0.75, 0.50, 0.50
pancreas	2.75/4	1.00, 0.75, 0.50, 0.50
common_class	2.75/4	1.00, 0.75, 0.50, 0.50
prototype	2.75/4	1.00, 0.75, 0.50, 0.50
baron	2.75/4	1.00, 0.75, 0.50, 0.50
segerstolpe	2.75/4	1.00, 0.75, 0.50, 0.50

Key/value reuse across recursions. Whether the first recursion's attention keys/values can be reused across later steps (a cache) instead of recomputed, trading a little accuracy for compute. (Table 12).

Warm-starting recursion from a fixed-depth model. Whether a recursive model benefits from initialising its shared block from a trained fixed-depth model and continuing training, versus training from scratch. (Table 13).

Table 13: Initialising the shared block from a trained fixed-depth model vs training from scratch.

Warm-start (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
fixed-depth source	75.2	51.8	69.0	92.9	68.1	8.3
MoR from scratch	71.7	62.8	65.8	95.5	61.5	5.4
MoR warm-started	65.8	61.6	64.8	93.3	55.8	6.9
warm-start gain	-5.9	-1.2	-0.9	-2.2	-5.6	+1.5

Routing configurations. How the routing behaves under different router heads (linear vs MLP), temperatures and load-balancing strengths, for both expert- and token-choice routing. (Table 14).

Table 14: Router head, temperature and load balancing for the expert- and token-choice routers.

Routing config (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe
expert linear	71.9	56.7	68.6	93.3	60.3	5.9
expert MLP	67.2	51.9	70.2	93.2	49.4	7.1
expert temp=2	73.5	57.6	66.7	94.1	66.2	6.5
token linear	74.7	53.7	70.1	94.5	52.6	5.4
token MLP	78.8	59.7	64.4	94.0	60.9	5.5
token balance=0.01	75.6	57.6	67.5	93.7	58.6	5.0

Visual summary. The same trends are shown graphically – scaling behaviour, adaptive recursion depth and parame-

Table 10: Cycle, Sequence, Middle-Cycle and Middle-Sequence sharing between the fully-shared and fully-independent extremes.

Sharing scheme (K=6)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe	prostate	blca	stad	panmeta_subtype
shared (1 block)	75.1	61.0	68.9	93.6	60.2	3.4	72.3	42.7	38.8	–
Cycle (3)	80.5	54.4	71.6	95.2	70.5	6.9	57.1	40.4	–	–
Sequence (3)	78.5	57.6	0.8	93.9	66.1	4.5	59.5	43.8	–	–
Middle-Cycle (3)	80.1	56.4	67.0	94.9	61.0	7.8	60.5	40.4	–	–
Middle-Sequence (3)	80.1	56.4	67.0	94.9	61.0	7.8	60.5	40.4	–	–
independent (6)	79.9	57.0	3.0	95.4	65.3	7.2	59.6	44.4	56.2	–

Table 12: Recomputing vs reusing the first-step attention keys/values across recursion steps.

KV strategy (macro-F1)	tabula_muris	pancreas	common_class	prototype	baron	segerstolpe	prostate	blca	stad	panmeta_subtype
recompute K/V (no cache)	79.9	54.1	70.4	93.9	65.6	8.6	74.7	43.3	36.4	–
reuse step-1 K/V (step-cache)	71.6	58.5	71.8	93.2	59.6	6.4	82.8	39.6	38.8	–

ter efficiency across the genomap suite (Fig. 3, Fig. 4 and Fig. 5).

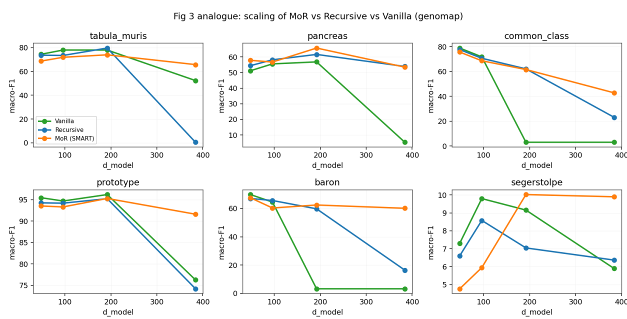

Figure 3: Macro-F1 versus model size for independent (Vanilla), weight-shared (Recursive) and adaptively-routed (SMART) stacks on each dataset.

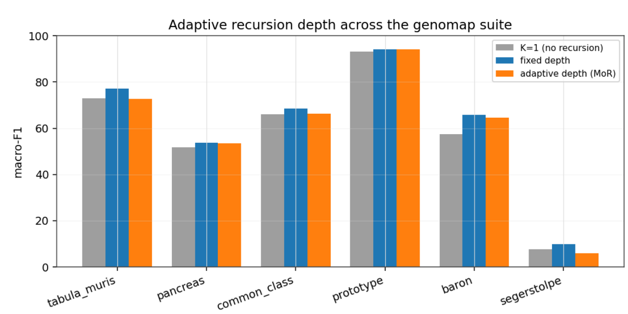

Figure 4: Adaptive recursion depth across all six genomap datasets: macro-F1 under no recursion (K=1), fixed depth, and adaptive routed depth.

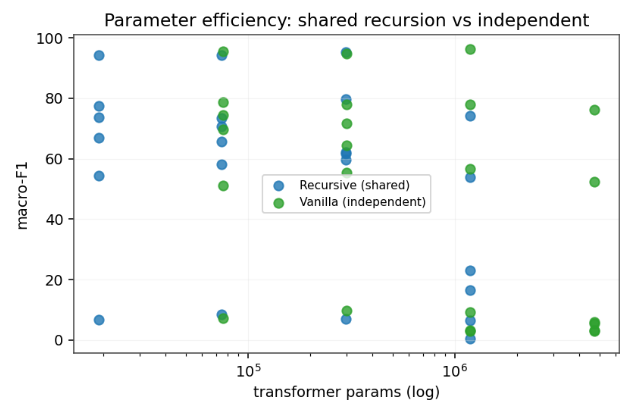

Figure 5: Accuracy versus transformer parameters for weight-shared recursion against independent layers.

6 Discussion and Limitations

SMART shows that biological inductive bias and parameter-efficient recursion can be co-designed: the same mechanism that makes the model small (weight sharing, compression) also makes it interpretable (markers, recursion depth), and a label-free gene-gene interaction prior can be folded directly into the routing decision without leaking labels or adding parameters. We scope the claims to what the evidence supports. (i) *The biology-informed prior is evaluated honestly.* Its benefit is established only to the extent that the co-expression graph separates from a degree-matched random-graph control (Sec. 4); we report that comparison as the runs deliver it and do not promote an effect the controls do not support. (ii) *Input format matters.* On the genomap-image pancreas inputs the router selects over pixels rather than genes, so its marker inductive bias does not fully apply and the macro-F1 drops; the gene-panel Tabula Muris benchmark is the on-distribution test. (iii) *Adaptive routing buys compute, not accuracy.* On these datasets the routing variants cluster within run-to-run noise, so the architecture's benefit is efficiency and the interpretable depth signal rather than an accuracy gain from depth itself. (iv) *Richer priors and broader data.*

Pathway-membership or regulatory-network centrality priors, the optional logit-Laplacian smoothing of Appendix C, and larger single-cell atlases are the natural next steps to test the prior where it should bite hardest.

7 Conclusion

We presented SMART, a recursive marker-guided transformer whose central novelty is a *biology-informed router*: a label-free genomap gene-gene interaction prior folded into a Mixture-of-Recursions depth decision, so biology shapes where the model spends computation rather than only how its results are read. By learning marker genes, compressing around them, and sharing one block across recursive refinement, SMART classifies single-cell types on Tabula Muris and a pancreas atlas with several times fewer transformer parameters than independent layers and an interpretable, compute-allocated recursion-depth signal. We evaluate the prior with a controlled none / co-expression / random-graph ablation and report the outcome transparently. The complete pipeline, including all experiments and this paper, regenerates from a single command.

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A Dataset Details

We use the two genomap single-cell capsule datasets (Islam and Xing 2023), converted to plain CSV (expression + labels + shipped split), summarised in Table 15. **Tabula Muris** is a fine-grained mouse cell atlas with a 1,089-feature genomap representation and its shipped 70/30 split. **Pancreas** is a human pancreatic integration atlas whose features are flattened 44×44 genomap images and which ships an integration train/test split. We deliberately restrict to these two datasets so that every table in the paper rests on the same single-cell, genomap-native setting.

B Effective-FLOPs Accounting

The compute numbers report per-sample FLOPs of the recursive transformer stack, the only component routing changes. One application of the shared block to a tokens costs $\phi(a) = 4a^2d + 4\, a\, d\, d_{\mathrm{ff}}$; the nominal fixed-depth cost is

Dataset	Cells	Features	Classes	Split
TM	54,865	1,089 genomap feats	55	70/30 (shipped)
Panc.	14,767	1,936 genomap feats	15	integration (shipped)
Common	27,499	1,089 genomap feats	19	stratified
Proto.	90,579	752 genomap feats	10	stratified
Baron	8,569	1,936 genomap feats	14	stratified
Seger.	2,122	1,936 genomap feats	10	stratified

Table 15: The two single-cell datasets used. Counts and splits are read directly from the data.

$\Phi_{\mathrm{nom}} = K\, \phi(M)$ and the effective cost sums one block over the tokens active at each step, $\Phi_{\mathrm{eff}} = \sum_{t=1}^{K} \phi(a_t)$, where a_t is the mean number of markers the expert-choice funnel keeps at step t. Gene embedding and marker selection are $\mathcal{O}(Nd)$, identical across routing modes, and excluded from this stack-level comparison.

C Theoretical Foundation of the Router

This appendix gives the mathematical and biological grounding for SMART's biology-informed router.

Routing as conditional computation. The router implements *conditional computation*: a learned policy routes each token to a token-specific amount of compute, the gating principle of sparsely-gated mixtures of experts (Shazeer et al. 2017), reused by Mixture-of-Depths (Raposo et al. 2024) and Mixture-of-Recursions (Bae et al. 2025); the "experts" here are recursion *depths* of one shared block, which couples adaptive computation (Graves 2016) to weight sharing.

The differentiable handle. The discrete top-k has zero gradient almost everywhere, so SMART keeps the soft probability of the chosen route as a multiplicative *gate* on the block output, $\mathbf{o}_m = g_m\, f_\theta(\mathbf{h}_m)$ with $g_m = \sigma(\tilde{r}_m)$. Because g_m is smooth in $\mathbf{w}_r$, the chain $\mathcal{L} \leftarrow \mathbf{o}_m \leftarrow g_m \leftarrow \mathbf{w}_r$ is unbroken: the hard choice routes, the soft gate carries the gradient. The biological prior of Eq. (1) is an additive constant in this logit, so it shifts the decision without breaking this path.

Biological foundation of the prior. Two facts from single-cell biology motivate the prior. A small set of *marker* genes carries most discriminative signal (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023), so compute should be allocated unevenly. And gene co-expression networks are approximately scale-free: a few high-degree *hub* genes (master regulators) exert outsized influence. We operationalise "hub" as eigenvector centrality on the genomap co-expression graph $\mathbf{W}$: the leading eigenvector $\mathbf{v}$ of $\mathbf{W}\mathbf{v} = \lambda\mathbf{v}$ scores each gene by the centrality of its neighbours, recursively, and we z-score it to form π.

Empirical-Bayes reading. Equation (1) is a log-linear prior on the routing decision, with $\beta_t\, \pi_m$ a Gaussian-like prior mean and $\beta_t = \beta_0(1 - \text{progress})$ a shrinkage strength that decays as data evidence accumulates: prior-dominated when the likelihood is uninformative (early training), data-dominated later.

Leakage safety. $\mathbf{W}$ is computed from training-split *expression only*; no cell-type label enters it. The prior therefore injects network topology, not the answer, which is why a gene-discovery claim under this prior is not circular, unlike a prior built from curated cell-type marker lists.

Optional pathway-graph smoothing. The additive bias treats genes independently. Co-pathway genes should route coherently, which one obtains by smoothing the logits over $\mathbf{W}$ with the normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, $\hat{\mathbf{r}}^{(t)} = \tilde{\mathbf{r}}^{(t)} - \gamma\,\mathbf{L}\tilde{\mathbf{r}}^{(t)}$ (graph-Laplacian regularisation): a gene borrows routing strength from its network neighbours. This is one sparse matrix-vector product per step and adds no transformer parameters; we expose it as an option and leave its evaluation to future work.